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MILITARY SPECIFICATION

FUZE, IMPACT, ATOMIC WEAPON: M1131

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type of impact fuze; designated Fuze, Impact, Atomic Weapon: M1131; which produces voltage pulses when subjected to impact shocks. This fuze is mounted in the tip of the inner cone assembly and is a component of the M35 impact fuze set which is used in the Pershing warhead section.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1135

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8805713 - Spring, Conical.
8861966 - Drum, Metal, Shipping, for Fuze, Impact: M1131 or M1136.
9321236 - Commercial Packaging, Packing, and Marking.
10561686 - Cover.
10561688 - Fuze, Impact, Atomic Weapon: M1131.
10561697 - Connector Assembly.
10561698 - Body.

PUBLICATIONS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

ET-10561688 - Inspection Equipment Tabulation.
QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Materials and components.- Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.2)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly.- The impact fuze shall comply with all requirements specified on drawing (dwg.) 10561688, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article.- When specified, the contractor shall furnish sample impact fuzes for first article inspection and approval (see 4.3 and 6.1.1).

3.4 Capacitance.- The capacitance of the impact fuze shall be as specified in Table I while at a temperature of 77° ± 2°F. In addition, the actual capacitance value in picofarads (pf) for each fuze output circuit shall be recorded (see 3.14). If

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the fuze fails to comply with the capacitance requirement after any environmental test (high temperature, temperature shock, low temperature, humidity, and vibration), the fuze shall be retested after being stabilized at 77° ± 2°F for a period of 24 hours. Testing shall be as specified in 4.5.1.

TABLE I

Output Circuit	Limits (pf)	Test Identification Number (see 3.14) 1/
J1	104 - 118	040001
J2	104 - 118	040002

1/ Not required during the following tests: high temperature, temperature shock, low temperature, humidity, and vibration.

* 3.5 Output voltage.- The peak output voltage of the impact fuze, as measured during the initial pulse, shall be as specified in Table II and shall be of positive polarity. The impact pulse applied to the fuze shall be 532 ± 27 gravities (g), peak amplitude, which will result in a maximum duration of the initial pulse of 200 microseconds as measured along the zero-voltage (reference) baseline on the measuring oscilloscope. If the fuze does not comply with the output voltage requirement of Table II, line (a) or (b), an alternate impact pulse of 650 ± 25 g shall be applied to the fuze, and the fuze shall be retested using the same test conditions that were used with the initial test. In addition, the minimum output voltage value (out of a total of three values to be measured from each of the two output circuits) shall be recorded (see 3.14) only for the acceptable g level. Testing shall be as specified in 4.5.2.

TABLE II

Input Pulse	Output Circuit	Peak Voltage	Test Identification Number (see 3.14) 1/
(a) 532 g	J1	3.0 min.	060001
(b) 532 g	J2	3.0 "	060002
(c) 650 g	J1	3.0 min.	060003
(d) 650 g	J2	3.0 "	060004

1/ Not required during the following tests: high temperature, temperature shock, low temperature, humidity, and vibration.

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3.6 Insulation resistance.- The insulation resistance of the impact fuze shall be not less than 1000 megohms. After environmental testing (high temperature, temperature shock, low temperature, and humidity), the insulation resistance of the fuze shall be not less than 500 megohms. There shall be no evidence of insulation breakdown or other damage during this test. Testing shall be as specified in 4.5.3.

* 3.7 Sensitivity.- The sensitivity of the impact fuze shall be as specified in Table III when subjected to a continuous sinusoidal vibration of 10 g ± 10 percent peak-to-peak amplitude at a frequency of 500 hertz ± 10 percent. In addition, the actual sensitivity value in coulombs/g, for each fuze output circuit, shall be recorded (see 3.14). Testing shall be as specified in 4.5.4.

TABLE III

Output Circuit	Sensitivity (coulombs/g)	Test Identification Number (see 3.14) 1/
J1	1×10^{-10} min.	070001
J2	1×10^{-10} "	070002

1/ Not required during the following tests: high temperature, temperature shock, low temperature, humidity, and vibration.

3.8 High temperature.- The impact fuze shall withstand a high temperature of 160° ± 5°F, for a minimum period of 4 hours, without evidence of deterioration or damage that may impair its intended operation. While at the high temperature (at the end of the high-temperature exposure period), the fuze shall comply with the requirements of 3.5. After this test (at room temperature), the fuze shall comply with the requirements of 3.4, 3.5, 3.6, and 3.7. Testing shall be as specified in 4.5.5.

3.9 Temperature shock.- The impact fuze shall withstand alternate exposures to temperatures of 160° ± 5°F and -65° ± 5°F, for a minimum period of 4 hours at each exposure, without evidence of deterioration or damage that may impair its intended operation. After this test (at room temperature), the fuze shall comply with the requirements of 3.4, 3.5, 3.6, and 3.7. Testing shall be as specified in 4.5.6.

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3.10 Low temperature.- The impact fuze shall withstand a low temperature of -33° ± 3°F, for a minimum period of 4 hours, without evidence of deterioration or damage that may impair its intended operation. While at the low temperature (at the end of the low-temperature exposure period), the fuze shall comply with the requirements of 3.5. After this test (at room temperature), the fuze shall comply with the requirements of 3.4, 3.5, 3.6, and 3.7. Testing shall be as specified in 4.5.7.

* 3.11 Humidity.- The impact fuze shall withstand a maximum relative humidity of 95 ± 5 percent and a maximum temperature of 130° ± 5°F without evidence of deterioration or damage that may impair its intended operation. After this test (at room temperature), the fuze shall comply with the requirements of 3.4, 3.5, 3.6, and 3.7. Testing shall be as specified in 4.5.8.

* 3.12 Vibration.- The impact fuze shall withstand vibrations as specified in Table IV without evidence of deterioration or damage that may impair its intended operation. During this test, the peak output voltage of the fuze shall not exceed 0.10 volt. After this test, the fuze shall comply with the requirements of 3.4, 3.5, 3.6, and 3.7. Testing shall be as specified in 4.5.9.

TABLE IV

Frequency Range (hertz)	Displacement (D.A.) or Peak g Level (± 10% Tol.)	Minimum Time Period to Cover Freq. Range (minutes)
10-30	0.20 inch	1.75
30-1000	8 g	2.0
1000-2000	10 g	3.75

3.13 Pull test on conical spring.- The conical spring shall be free from embrittlement, after plating, as evidenced by not breaking or cracking when extended to the point of minimum permanent deformation. Testing shall be as specified in 4.5.10.

3.14 Serialization.- A six-letter serial code shall be assigned to each serialized production item. The contractor shall obtain these serial codes by means of a letter addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAN-R, Dover, New Jersey 07801. The letter of request shall include: (a) exact nomenclature of the item as specified on the applicable drawing (see dwg. 10561688), (b) contract number or procurement request order

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number, and (c) number of serial codes required (unused codes must be returned). In addition to the normal inspection records, the contractor shall provide data feedback (see 6.1.2) in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the test identification numbers, are specified in 3.4, 3.5, and 3.7.

* 3.15 Soldering.- The connector assembly (dwg. 10561697) shall be soldered in accordance with MIL-STD-1460.

3.16 Workmanship.- All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection.

* 4.3.1 Submission.- When the contract requires first article inspection, the contractor shall submit the following to the testing facility designated in the invitation for bids or request for proposal: (a) three consecutively produced impact fuzes and (b) three impact fuze bodies (dwg. 10561698)(for defect inspection (see 4.4.2.1.1)).

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* 4.3.2 Inspection and testing to be performed.- The first article inspection samples shall be inspected, as applicable, for the defects specified in 4.4.2.1.1 and for the defects specified in 4.4.2.1.2 (except defects 103 and 104) using inspection methods contained therein, and for all requirements of Section 3, which are listed in Table V, using the test methods specified in 4.5. Inspection and testing shall be performed with the acceptance inspection equipment specified in 4.4.4. The impact fuzes shall be tested in the order specified in Table V. If any impact fuze, or fuze body, fails to comply with any of the applicable requirements, the first article samples shall be rejected. The Government reserves the right to terminate its inspections upon any failure of the first article samples to comply with the stated requirements. This series of tests is considered destructive.

TABLE V. First article tests.

Test	Requirement paragraph
1. Capacitance	3.4
2. Output voltage	3.5
3. Insulation resistance	3.6
4. Sensitivity	3.7
5. High temperature	3.8
6. Temperature shock	3.9
7. Low temperature	3.10
8. Humidity	3.11
9. Vibration	3.12

4.4 Quality conformance inspection.

4.4.1 Inspection lot.- A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination.- A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table VI). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

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TABLE VI. Inspection lot sampling.

Major Defect Table		Minor Defect Table	
Lot Size	Sample Size	Lot Size	Sample Size
1-30	all	1-30	all
31-45	29	31-45	27
46-60	35	46-60	30
61-75	38	61-75	33
76-95	42	76-95	35
96-130	46	96-130	37
131-180	50	131-180	39
181-250	52	181-250	41
251-400	54	251-400	42
401 and over	56	401 and over	44

4.4.2.1 Classification of defects.- Critical, major, and minor defects are defined in MIL-STD-105.

4.4.2.1.1 Body (see dwg. 10561698). 1/

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of tapped hole in smaller end	Gage
102.	Minor diameter of tapped hole in smaller end, max	Gage
103.	Length to end of threads of hole in smaller end, min	Gage
104.	Length to shoulder of hole in smaller end, min	Gage
105.	Diameter of counterbore in smaller end, min	Gage
106.	Concentricity of pitch diameter of threaded hole in smaller end with tapered surface at larger end	Gage
107.	Concentricity of counterbore in smaller end with tapped portion of same hole	Gage
108.	Outside diameter at smaller end, max	Gage
109.	Concentricity of outside diameter at smaller end with tapered surface at larger end	Gage
110.	Angularity of tapered surface at larger end with flat surface at larger end	Gage

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111. Location of tapped hole in tapered surface near larger end.....Gage
112. Pitch diameter of tapped hole in tapered surface near larger end.....Gage
113. Depth of thread in tapped hole in tapered surface near larger end, min.....Gage
114. Pitch diameter of tapped hole in flat surface at larger end, max. (4 holes).....Gage
115. Minor diameter of tapped hole in flat surface at larger end, max. (4 holes).....Gage

Minor:

201. Length, max.....Gage
202. Diameter of counterbore in tapered surface near larger end, min.....Gage
203. Depth of counterbore in tapered surface near larger end, min.....Gage
204. Length from smaller end to beginning of tapered surface, min.....Gage
205. Step-down in tapered surface missing or improper.....Visual
206. Evidence of poor workmanship (see 3.16).....Visual

1/ The three impact fuze bodies, submitted as part of the first article samples, shall be used for this inspection during first article testing.

* 4.3.2.1.2 Fuze, Impact, Atomic Weapon: M113l (see dwg. 10561688).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Torque on cover mounting screw, min. (4 screws)	Gage
102.	Ability of connector to engage and disengage with mating receptacle (2 connectors)	Manual/Gage
103.	Bonding material missing or improper (between cover and body and between connector assembly and cover (2 places)) (inspect during assembly)	Visual
104.	Soldering of connector assembly improper (inspect during assembly) (see 3.16)	Visual

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Minor:

201.	Weight	Approved weighing device
202.	Connector insert damaged (2 places)	Visual
203.	Protective cap, on connector assembly, missing or improper (2 places)	Visual
204.	Marking missing, improper, or illegible	Visual
205.	Protective finish missing or improper (see dwg. 10561686 and 10561698)	Visual
206.	Evidence of poor workmanship (see 3.16)	Visual

* 4.4.2.1.3 Drum, Metal, Shipping, for Fuze, Impact: M1131 or M1136 (prior to securing cover)(see dwg. 8861966).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Impact fuze improperly wrapped and bagged	Visual/Manual
102.	Cushioning disk (2 places) and cutout disk improperly assembled	Visual/Manual
Minor:		
201.	Bag marking missing, improper, or illegible	Visual
202.	Evidence of poor workmanship (see 3.16)	Visual

* 4.4.2.1.4 Drum, Metal, Shipping, for Fuze, Impact: M1131 or M1136 (see dwg. 8861966).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Drum improperly sealed (including metallic seal)	Visual/Manual
102.	Drum damaged (including punctures, dents, and gouges)	Visual

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Minor:

201. Scratch on drum exposing base metal.....Visual
202. Marking missing, improper, or illegible.....Visual
203. Evidence of poor workmanship (see 3.16).....Visual

* 4.4.2.1.5 Packaging of Impact Fuzes for Interplant Shipments (see 5.1 and 6.2.2).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence available (see 6.2.1) that impact fuzes were preserved, packaged, packed, and marked in accordance with dwg. 9321236 (see 5.1).....	Visual
Minor:	None defined.	

4.4.3 Inspection testing.- The contractor shall perform all of the tests specified in Table VII. The tests on the impact fuze (tests 1 to 6) shall be performed in the listed sequence. Of these tests, No. 7 is considered destructive and samples so tested shall not be returned to the lot.

* TABLE VII. Quality conformance tests.

Test	Test Classification
1. Capacitance	Major
2. Output voltage	Major
3. Insulation resistance	Major
4. Sensitivity	Major
5. High temperature	Major
6. Low temperature	Major
7. Pull test on conical spring	Major
8. Pull test on connector assembly	Minor

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* 4.4.3.1 Capacitance (see 3.4), output voltage (see 3.5), insulation resistance (see 3.6), and sensitivity (see 3.7).- Each impact fuze shall be subjected to these tests (see applicable paragraph in 4.5). If any fuze fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.4.3.2 High temperature (see 3.8) and low temperature (see 3.10).- Two impact fuzes shall be selected at random from every lot of 100 fuzes produced, or portion thereof, and subjected to these tests (see applicable paragraph in 4.5). If any fuze fails to comply with the applicable requirements, the lot shall be rejected.

4.4.3.3 Pull test on connector assembly (see dwg. 10561697).- Each connector assembly shall be subjected to this test (see 4.5.11). If any connector assembly fails to comply with the pull test requirement of the drawing, it shall be classified defective and be removed from the lot.

4.4.3.4 Pull test on conical spring (see 3.13 and dwg. 8805-713).- Ten conical springs shall be selected from each heat-treat lot of springs submitted for acceptance, and they shall be subjected to this test (see 4.5.10). If any spring fails to comply with the applicable requirements, the lot of springs, represented by the samples where the failure occurred, shall be rejected.

4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests.- Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions as specified in the Equipment List (EL) listed in the Equipment Tabulation, ET-10561688. When either of the following conditions exists, the contractor shall design required equipment or specify suitable commercial equipment: (a) if any deviation from equipment or procedures, as specified in the EL, is desired or (b) the code "CD" is listed in the "Part or Identifying Number" column in the EL (see 6.1.2). The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions prior to use on the contract (see 6.3).

4.4.4.2 Minor defects and tests.- Acceptance inspection equipment for minor defects and tests is not specified in ELs. The acceptance inspection equipment, operating instructions, and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative responsible for acceptance inspection.

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4.5 Methods of inspection.- The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F). NOTE: During environmental tests (see Table V, tests 5 to 8), the impact fuze connectors shall be protected by their protective caps.

4.5.1 Capacitance.- The capacitance of the impact fuze shall be measured by using a Government-approved capacitance bridge at a frequency of 1000 hertz. The capacitance measurement shall be made between the center terminal and the shell of each connector, and the fuze shall comply with the requirements of 3.4 during this test.

* 4.5.2 Output voltage.- The impact fuze shall be mounted to a Government-approved test fixture with the fuze cover plate in the upper position. The fuze shall be assembled to the test fixture using the threaded hole in the narrow end of the fuze. The fuze shall be tightened to the test fixture with a torque of 19 ± 2 foot-pounds (ft-lb). The output voltage of the fuze shall be measured across a fixed load consisting of a 1000 ohm resistor (± 1 percent) shunted by a 0.1 microfarad capacitor (± 2 percent). A Government-approved, dual-trace oscilloscope shall be used to simultaneously measure the output voltage pulse and the impact shock. The fuze shall be impacted three times along a vertical axis which is perpendicular to the fuze cover plate, and the output voltage from each connector shall be measured simultaneously. The fuze shall comply with the requirements of 3.5 during this test.

4.5.3 Insulation resistance.- The insulation resistance of the impact fuze shall be measured by using a Government-approved insulation resistance tester at an applied voltage of 1600 ± 100 volts direct current. The measurements shall be made between the center terminal and the shell of each connector assembly with the positive test voltage applied to the center terminal. The test voltage shall be applied instantaneously to the fuze, and the insulation resistance shall be measured after a 10-second application of this test voltage. The fuze shall comply with the requirements of 3.6 during this test.

* 4.5.4 Sensitivity.- The impact fuze shall be mounted to a vibration machine with the fuze cover plate in the upper position. The fuze shall be assembled to the vibration machine using the threaded hole in the narrow end of the fuze. The fuze shall be tightened to the vibration machine with a torque of 19 ± 2 ft-lb. The fuze shall be subjected to the vibrations specified in 3.7. The fuze shall be vibrated

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along a vertical axis which is perpendicular to the fuze cover plate. The output of the impact fuze shall be fed directly to a charge amplifier or to a suitable high impedance rms voltmeter (taking into account the system capacitance in converting to coulombs/g when using the voltmeter). The measurements from output connectors J1 and J2 do not have to be taken simultaneously, but may be taken separately. The duration of this test is not significant, but it shall be long enough to ensure obtaining valid measurements. The fuze shall comply with the requirements of 3.7 during this test.

4.5.5 High temperature.- The impact fuze shall be placed into a test chamber, and the temperature of the chamber shall be raised to 160°F. The fuze shall remain at this temperature for a minimum period of 4 hours. At the end of this exposure period, the fuze shall be removed from the test chamber and, within a 3-minute period, it shall be tested for the requirements of 3.5. The fuze shall then be allowed to stabilize at room temperature, and it shall then be tested for the applicable requirements of 3.4, 3.5, 3.6, and 3.7. The fuze shall comply with the requirements of 3.8 during this test.

4.5.6 Temperature shock.- The impact fuze shall be placed into a test chamber having an internal temperature of 160°F, and it shall remain at this temperature for a minimum period of 4 hours. At the end of this test exposure, and within a 3-minute period, the fuze shall be transferred to a test chamber having an internal temperature of -65°F, and it shall remain at this temperature for a minimum period of 4 hours. This high- and low-temperature exposure constitutes one cycle, and the fuze shall be subjected to three such cycles. The duration of exposure at each extreme temperature shall be not less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the completion of the third cycle (at the end of the low-temperature exposure period), the fuze shall be removed from the test chamber and be allowed to stabilize at room temperature. The fuze shall then comply with the requirements of 3.9.

4.5.7 Low temperature.- The impact fuze shall be placed into a test chamber, and the temperature of the chamber shall be lowered to -33°F. The fuze shall remain at this temperature for a minimum period of 4 hours. At the end of this exposure period, the fuze shall be removed from the test chamber and, within a 3-minute period, it shall be tested for the requirements of 3.5. The fuze shall then be allowed to stabilize at room temperature, and it shall then be tested for the applicable requirements of 3.4, 3.5, 3.6, and 3.7. The fuze shall comply with the requirements of 3.10 during this test.

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* 4.5.8 Humidity.- The impact fuze shall be subjected to the following humidity test:

(a) The fuze shall be placed into a dry oven and conditioned at a temperature of 130°F for a minimum period of 24 hours.

(b) The fuze shall be transferred to a humidity chamber having a temperature of 75° ± 5°F and a relative humidity of 50 ± 10 percent. The fuze shall remain in this chamber for a minimum period of 24 hours.

(c) The temperature of the chamber shall be gradually raised, within a 2-hour period, to 105° ± 5°F and a relative humidity of 95 percent. The fuze shall remain in this chamber for a minimum period of 16 hours.

(d) The temperature of the chamber shall be gradually decreased, within a 2-hour period, to 70° ± 5°F, and the relative humidity shall be maintained at 95 percent. The fuze shall remain in this chamber for a minimum period of 4 hours.

(e) Repeat steps (c) and (d) for a total of 10 cycles (240 hours).

(f) At the end of the last cycle of this test, the fuze shall be removed from the test chamber, and it shall comply with the requirements of 3.11. NOTE: Accumulated moisture (if any), on the fuze, may be removed by gentle use of dry compressed air.

* 4.5.9 Vibration.- The impact fuze shall be mounted to a vibration machine with the fuze cover plate in the upper position. The fuze shall be assembled to the vibration machine using the threaded hole in the narrow end of the fuze. The fuze shall be tightened to the vibration machine with a torque of 19 ± 2 ft-lb. The fuze shall be subjected to the vibrations specified in Table IV along a vertical axis which is perpendicular to the fuze cover plate. During this test, the output voltage of each connector assembly shall be monitored using the same method (and load) as specified in 4.5.2. The fuze shall comply with the requirements of 3.12 during this test.

4.5.10 Pull test on conical spring.- The conical spring shall be attached to a Government-approved test fixture which will securely clamp approximately one half of the coil at the large end of the spring. The coil at the small end of the spring shall be pulled so that the spring is extended, relaxed, and

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re-extended to greater lengths until deformation occurs. The spring shall comply with the requirements of 3.13 during this test.

4.5.11 Pull test on connector assembly.- The connector assembly shall be connected to a Government-approved test fixture which will be capable of exerting a pull force (specified on dwg. 10561697) between the receptacle connector and the connecting wafer. The pull force shall be applied for a minimum period of 60 seconds. After this test, the connector assembly shall be inspected for evidence of damage or separation of the two soldered joints.

* 5. PACKAGING

5.1 Impact fuzes for interplant shipments (see 6.2.2).

5.1.1 Preservation, packaging, packing, and marking.- Preservation, packaging, packing, and marking shall be in accordance with dwg. 9321236 or as specified in the contract or purchase order.

5.2 Impact fuzes for spare parts.

5.2.1 Preservation, packaging, and marking.- Preservation, packaging, and marking shall be in accordance with dwg. 8861966.

5.2.2 Packing and marking.- Packing and marking shall be in accordance with dwg. 9321236 or as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

* 6.1.1 Procurement requirements.- (a) Title, number, and date of this specification, (b) whether procurement is for fuzes for spare parts or for interplant shipments, (c) selection of methods of packaging or packing (see 5.1 and 5.2), and (d) first article sample requirements.

6.1.2 Data requirements.- Data specified in 3.4, 3.5, and 3.7, and contractor engineering documentation specified in 4.4.4.1, will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.

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* 6.2 Definitions.

6.2.1 Objective evidence.- Records of the contractor's quality control and inspections which can be verified.

6.2.2 Interplant shipments.- Shipments from the supply source to a receiving activity for subsequent incorporation into the next assembly during end item production.

6.3 Acceptance inspection equipment.- The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAN-I, Dover, New Jersey 07801.

* 6.4 Drawings and publications.- Drawings and publications listed in Section 2 of this specification under the heading of US Army Armament Research and Development Command (ARRADCOM) may also include drawings and publications prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings and publications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes from previous issue.- The margins of this specification are marked with asterisks to indicate where major changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-AR

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